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1.1 👤 You

CM2526_exercises_09

PDF

Please solve the given exercise sheet. Give all exercise with detailed solution in LaTeX syntax.

1.2 🤖 Gemini

Show thinking

Here are the solutions to the exercises from the provided condensed matter physics sheet.

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Exercise 1: Wave equation in continuum

Problem Statement: Show that the N coupled equations of motion for a mono-atomic chain reduce to the continuum elastic wave equation for long wavelengths.

Solution: The equation of motion for the n-th atom is given by:

$Mdt2d2undt2dt2t2d2und2d2unun = ppCpCpun + pun + pn + p - unun$

where $unun$ is the displacement of atom n and $CpCp$ are the force constants. For a mono-atomic chain, we assume symmetry such that $CpCp = C - pC - p - p$.

1. Continuum Approximation For long wavelengths ($\lambda \gg a$), the displacement $ununt$ changes slowly from atom to atom. We can treat the displacement as a continuous function uxt, where $x = na$. We expand $un + pun + pn + p = ux + pa$ in a Taylor series around x:

$ux + pa \approx ux + \partial x\partial u\partial x\partial x\partial u\partial upa + 212211\partial x2\partial 2u\partial x2\partial x2x2\partial 2u\partial 2\partial 2upa2+$

Substituting this into the term $un + pun + pn + p - unun$:

$un+pun+pn+p-unun \approx pa\partial x\partial u\partial x\partial x\partial u\partial u+212211p2p2a2a2\partial x2\partial 2u\partial x2\partial x2x2\partial 2u\partial 2\partial 2u$

2. Summation Substitute this back into the equation of motion:

$M\partial t2\partial 2u\partial t2\partial t2t2\partial 2u\partial 2\partial 2u = pCpCppa\partial x\partial u\partial x\partial x\partial u\partial u+212211p2p2a2a2\partial x2\partial 2u\partial x2\partial x2x2\partial 2u$

Separating the terms:

$M\partial t2\partial 2u\partial t2\partial t2t2\partial 2u\partial 2\partial 2u = appCpCp\partial x\partial u\partial x\partial x\partial u\partial u+2a222a2a2a2pp2p2CpCp\partial x2\partial u\partial x2\partial x2$

3. Symmetry Argument Due to inversion symmetry of the Bravais lattice, the force constants satisfy $CpCp = C - pC - p - p$. Therefore, the term linear in p vanishes:

$ppCpCp = 0$

The equation reduces to:

$M\partial t2\partial 2u\partial t2\partial t2t2\partial 2u\partial 2\partial 2u = p212211p2p2a2a2CpCp\partial x2\partial 2u\partial x2\partial x2x2\partial 2u\partial 2\partial 2u$

4. Result Dividing by M, we obtain the continuum elastic wave equation:

$\partial t2\partial 2u\partial t2\partial t2t2\partial 2u\partial 2\partial 2u = vs2vs2\partial x2\partial 2u\partial x2\partial x2x2\partial 2u\partial 2\partial 2u$

where the speed of sound $vsvs$ is identified as:

$vsvs = M1p\sum 21p2a2CpM1p\sum 21p2a2CpM1MM11p212211p2p2a2a2CpCp$

Exercise 2: Linear chain with next nearest neighbour interactions

Problem Statement: A linear chain with nearest neighbor (NN) coupling $C1C1$ and next-nearest neighbor (NNN) coupling $C2C2$. $C2C2$ is ν times smaller than $C1C1$ ($C2C2 = C1C1/\nu$).

(a) Find the dispersion relation ωk The equation of motion including NN ($p = \pm 1$) and NNN ($p = \pm 2$) interactions is:

$Mu\ddot{}nu\ddot{}u\ddot{}n = C1C1un + 1un + 1n + 1 + un - 1un - 1n - 1 - 2unun + C2C2un + 2un + 2n + 2 + un - 2un - 2n - 2 - 2unun$

Using the ansatz $unun t = Aei(kna - \omega t)ei(kna - \omega t)ikna - \omega t$, we find:

$-M\omega 2\omega 2 = C1C1eikaeikaika + e - ikae - ika - ika - 2 + C2C2ei2kaei2kai2ka + e - i2kae - i2ka - i2ka - 2$

$M\omega 2\omega 2 = 2C1C11 - ka + 2C2C21 - 2ka$

Using the identity $1 - x = 22x/2$:

$\omega 2\omega 2k = M4C1MM4C14C1C122ka22kaka + M4C2MM4C24C2C22ka$

Substituting $C2C2 = C1C1/\nu$:

$\omega k = M4C1(sin2(2ka){+}\nu 1sin2(ka))M4C1(sin2(2ka){+}\nu 1sin2(ka))M4C1MM4C14C1C122ka22$
$\nu 1\nu\nu 112ka$

(b) For which integer values of ν does the maximum lie at $k < \pi/a$? To find the extrema, we take the derivative of $\omega 2\omega 2$ with respect to k and set it to zero:

$dkd(\omega 2)dkdkd(\omega 2)d\omega 2\omega 2 = M2C1aMM2C1a2C1C1aka{+}M2C2(2a)MM2C2(2a)2C2C22a2ka =$
0

Using $2ka = 2kaka$:

$M2aMM2a2akaC1C1 + 4C2C2ka = 0$

The solutions are $ka = 0$ (at $k = 0\pi/a$) or:

$ka = -4C2C14C24C2C2C1C1C1$

Substituting $C2C2 = C1C1/\nu$, the condition becomes:

$ka = -4(C1/\nu)C14(C1/\nu)4C1C1/\nu C1C1C1 = -4\nu 44\nu\nu$

For a maximum to exist inside the Brillouin zone ($0 < k < \pi/a$), we must have $\mid ka \mid < 1$. Thus:

$-4\nu 44\nu\nu < 1 \Longrightarrow \nu < 4$

Since ν is an integer (and typically $\nu \geq 1$ as $C2C2$ is "smaller"), the values are: $\nu = 123$

(c) Maximum frequency for $\nu = 2$ For $\nu = 2$, the maximum occurs at $ka = -2/4 = -1/2$. This corresponds to $ka = 2\pi/3$. Substitute this into the dispersion relation with $C2C2 = C1C1/2$:

$$\omega max2\omega maxmax2 = M2C1MM2C12C1C11 - -1/2 + M2(C1/2)MM2(C1/2)2C1C1/21 - 4\pi/3$$

Note that $4\pi/3 = -1/2$.

$$\omega max2\omega maxmax2 = M2C1MM2C12C1C1232233 + MC1MMC1C1C11 - -0.5$$

$$\omega max2\omega maxmax2 = M3C1MM3C13C1C1 + M1.5C1MM1.5C11.5C1C1 = M4.5C1MM4.5C14 2M9C12M2M9C19C1C1$$

$$\omega max\omega maxmax = 32MC12MC12MC12M2MC1C1C1$$

This is higher than the frequency at the boundary $\omega\pi/a = 4C1/M4C1/M4C1C1/M = 2C1/MC1/MC1C1/M$.

(d) Speed of sound $vsvs$ The speed of sound is the group velocity at $k \to 0$. For small k, $x \approx x$.

$$\omega 2\omega 2 \approx M4C1MM4C14C1C12ka22kaka2 + M4C2MM4C24C2C2ka2$$

$$\omega 2\omega 2 \approx MC1a2k2MMC1a2k2C1C1a2a2k2k2 + M4C2a2k2MM4C2a2k24C2C2a2a2k2k2 = Ma2k2MMa2k2a2a2k2k2C1C1 + 4C2C2$$

Taking the square root:

$$\omega \approx MC1 + 4C2MC1 + 4C2MC1 + 4C2MMC1 + 4C2C1C1 + 4C2C2ak$$

Thus,

$$vsvs = dkd\omega dkdkd\omega d\omega k \to 0k \to 0k \to 0 = aMC1 + 4C2MC1 + 4C2MC1 + 4C2MMC1 + 4C2C1C1 + 4C2C2$$

Exercise 3: Atomic chain of two types of atoms

Problem Statement: Diatomic chain, masses $M1M1$ and $M2M2$. $M1M1$ at $xnxn = na$, $M2M2$ at $xnxn + d$ ($d < a$). Nearest neighbor coupling D.

(a) Differential equations Let $u1, nu1, n1n$ be the displacement of mass $M1M1$ in cell n, and $u2, nu2, n2n$ for mass $M2M2$. The NN interactions for $M1M1$ are with $M2M2$ in the same cell (dist d) and $M2M2$ in the previous cell (dist $a - d$). Since the coupling constant is D for both:

$$M1M1\ddot{u}1, nu\ddot{u}\ddot{u}1, n1n = Du2, nu2, n2n - u1, nu1, n1n + Du2, n - 1u2, n - 12n - 1 - u1, nu1, n1n$$

$$M2M2\ddot{u}2, nu\ddot{u}\ddot{u}2, n2n = Du1, nu1, n1n - u2, nu2, n2n + Du1, n + 1u1, n + 11n + 1 - u2, nu2, n2n$$

(b) Dispersion relation and BZ Sketch Using the ansatz $M\alpha M\alpha M\alpha M\alpha u\alpha, nu\alpha, n\alpha n = A\alpha A\alpha ei(kna - \omega t)ei(kna - \omega t)ikna - \omega t$: Substitute into EOMs:

$$M1M1 - \omega 2\omega 2M1A1M1M1M1M1M1A1A1A1 = DM2A2M2M2M2M2M2A2A2A2 - M1A1M1M1M1M1M1A1A1A1 + DM2A2M2M2M2M2M2A2A2A2e - ikae - ika - ika - M1A1M1M1M1M1M1A1A1A1$$

$M2M2 - \omega 2\omega 2M2A2M2M2M2M2M2A2A2A2 = DM1A1M1M1M1M1M1A1A1A1 - M2A2M2M2M2M2M2A2A2A2 + DM1A1M1M1M1M1M1A1A1A1eikaeikaika - M2A2M2M2M2M2M2A2A2A2$

This simplifies to the linear system:

$M12DM1M1M12D2D - \omega 2\omega 2A1A1 - M1M2DM1M2M1M2M1M2M1M1M2M2DD1 + e - ikae - ika - ikaA2A2 = 0$

$-M1M2DM1M2M1M2M1M2M1M1M2M2DD1 + eikaeikaikaA1A1 + M22DM2M2M22D2D - \omega 2\omega 2A2A2 = 0$

Setting the determinant to zero:

$M12DM1M1M12D2D - \omega 2\omega 2M22DM2M2M22D2D - \omega 2\omega 2 - M1M2D2M1M2M1M1M2M2D2$
$1 + e - ikae - ika - ika \mid 2 \mid 2 = 0$

Using $\mid 1 + e - ikae - ika - ika \mid 2 \mid 2 = 2 + 2ka = 42ka/2 = 4 - 42ka/2$:

$\omega 4\omega 4 - 2DM11M1M1M111 + M21M2M2M211\omega 2\omega 2 + M1M24D2M1M2M1M1M2M24D24D2D$
0

Solving for $\omega 2\omega 2$:

$\omega 2\omega 2 = DM11M1M1M111 + M21M2M2M211(M11 + M21)2 - M1M24sin2(2ka)(M11 + M21)2 - M1M24sin2(2ka)M11M1M1M111 + M21M2M2M2112 - M1M24M1M2M1M1M2M2$

$+$ sign: Optical branch.

$-$ sign: Acoustic branch.

Sketch: The first Brillouin zone is $-\pi/a\pi/a$. The acoustic branch starts at 0, optical starts at a finite max value. They approach a gap at the zone boundary $\pm\pi/a$.

(c) Group velocity $vgvg = d\omega/dk$.

Acoustic: $vgvg$ is constant (sound speed) near $k = 0$, decreases to 0 at $k = \pm\pi/a$.

Optical: $vgvg$ is 0 at $k = 0$, increases, then decreases back to 0 at $k = \pm\pi/a$.

(d) Solutions for limits

1. $ka \ll 1$ (Center of zone)

Acoustic: $\omega \approx vsvsk$. $A1A1 \approx M1/M2M1/M2M1M1/M2M2A2A2$. More physically $u1u1 \approx u2u2$. The atoms in the cell move in phase (sound wave).

Optical: $\omega \approx 2D(1/M1 + 1/M2)2D(1/M1 + 1/M2)2D1/M1M1 + 1/M2M2$. $M1M1u1u1 + M2M2u2u2 \approx 0$. The atoms move out of phase against each other (center of mass stationary).

2. $ka = \pm\pi$ (Zone boundary) $2ka/2 = 1$. The term in the square root becomes $1/M1M1 - 1/M2M22$.

Optical: $\omega 2\omega 2 = 2D1/M1M1 + 1/M2M2 + 2D \mid 1/M1M1 - 1/M2M2 \mid$. Assume $M1M1 < M2M2$. Then $\omega 2\omega 2 = 2D1/M1M1 + 1/M2M2 + 1/M1M1 - 1/M2M2 = M14DM1M1M14D4D$.

Frequency depends on the lighter mass.

Motion: Heavier mass is stationary ($AheavyAheavyheavy = 0$), Lighter mass oscillates.

Acoustic: $\omega 2\omega 2 = 2D/M2M2$ (if $M1M1 < M2M2$).

Frequency depends on the heavier mass.

Motion: Lighter mass is stationary ($AlightAlightlight = 0$), Heavier mass oscillates.

(e) Case $M1M1 \gg M2M2$ If $M1M1$ is very heavy:

Optical Branch: $\omega \approx 2D/M22D/M22D/M2M2$. The light atoms ($M2M2$) oscillate rapidly while heavy atoms ($M1M1$) are effectively stationary walls. The band is flat (Einstein oscillator model).

Acoustic Branch: ω is small, governed by the total mass or the heavy mass near the boundary.

(f) Numerical Calculation for NaCl

$MNaMNaNa \approx 23$ u (let $M1M1$), $MClMClCl \approx 35.5$ u (let $M2M2$).

$1uu \approx 1.66 \times 10-270-27-27$ kg.

$D = 20$ N/m.

1. Width of Optical Branch ($\Delta\omega$)

$\omega opt\omega optopt0 = 2D(M11{+}M21)2D(M11{+}M21)2DM11M1M1M111{+}M21M2M2M211$

$\omega opt\omega optopt\pi/a = M12DM12DM12DM1M1M12D2DsincesinceM1M1 < M2M2$

Converting masses to kg: $M1M1 \approx 3.82 \times 10-260-26-26$ kg, $M2M2 \approx 5.89 \times 10-260-26-26$ kg.

$\omega opt\omega optopt0 = 40(3.821 + 5.891) \times 102640(3.821 + 5.891) \times 1026403.8213.823.8211 + 5.8915.895.8911 \times 102602626 \approx 4.15 \times 101301313 rad/srad/s$

$\omega opt\omega optopt\pi/a = 3.82 \times 10-26403.82 \times 10-26403.82 \times 10-26403.82 \times 10-263.82 \times 10-260-26-264040 \approx 3.24 \times 101301313 rad/srad/s$

$\Delta\omega opt\omega optopt = 4.15 - 3.24 \times 101301313 = 0.91 \times 101301313 rad/srad/s$

2. Energy Gap Gap is between min of optical and max of acoustic. Max acoustic is at boundary: $\omega ac\omega acac\pi/a = 2D/M22D/M22D/M2M2$.

$\omega ac\omega acac\pi/a = 5.89 \times 10-26405.89 \times 10-26405.89 \times 10-26405.89 \times 10-265.89 \times 10-260-26-264040 \approx 2.61 \times 101301313 rad/srad/s$

Frequency Gap $\Delta\omega gap\omega gapgap = 3.24 \times 101301313 - 2.61 \times 101301313 = 0.63 \times 101301313 rad/srad/s$. Energy Gap $EgEg = \hbar\Delta\omega gap\omega gapgap$.

$EgEg \approx 6.582 \times 10-160-16-16 eVeV \cdot ss \times 0.63 \times 101301313 s-1ss-1-1 \approx 4.1 meVmeV$

$FrequencyGapFrequencyGap \approx 6.3 \times 101201212 rad/srad/s$

Next Step: Would you like me to visualize the dispersion relation for Exercise 2 to verify the condition $\nu < 4$?

Sources